K-12 Grade-Banded Engineering Design Standards

The iterative cycle of design offers the opportunity for students to apply science knowledge in the classroom and engage in the engineering practices. The components of engineering design have been broken down by *a Framework for K-12 Science Education* and are listed, below:

Core Idea ETS1: Engineering Design

ETS1.A: Defining and Delimiting an Engineering Problem

ETS1.B: Developing a Possible Solution

ETS1.C: Optimizing the Design Solution

It is essential that every unit of instruction in science or engineering must include a goal for students to develop understanding of at least one disciplinary core idea in Life, Physical, or Earth/Space Science.

These standards are optional points of access for integrating the engineering design process into science lessons/units/curriculum. The standards are broken down by grade-band.

Note: *These standards will NOT be assessed on the statewide monitoring exam at grades 5, 8 and 11.*

K-2 Engineering Design Standards

Engineering design in the earliest grades introduces students to "problems" as situations that people want to change. They can use tools and materials to solve simple problems, use different representations to convey solutions, and compare different solutions to a problem and determine which is best. Students in all grade levels are not expected to come up with original solutions, although original solutions are always welcome. Emphasis is on thinking through the needs or goals that need to be met, and which solutions best meet those needs and goals.

Grades K-2 Engineering Design Standards	
K-2-ETS1-1.	Ask questions, make observations, and gather information about a situation people want to change to define a simple problem that can be solved through the development of a new or improved object or tool. *Alignment may include K-PS2-2, K-ESS3-2, 1-LS1-1*
K-2-ETS1-2.	Develop a simple sketch, drawing, or physical model to illustrate how the shape of an object helps it function as needed to solve a given problem. *Alignment may include K-ESS3-3, 1-PS4-4, 2-LS2-2*
K-2-ETS1-3.	Analyze data from tests of two objects designed to solve the same problem to compare the strengths and weaknesses of how each performs. *Alignment may include 2-PS1-2, 2-ESS2-1*

3-5 Engineering Design Standards

At the upper elementary grades, engineering design engages students in more formalized problem solving. Students define a problem using criteria for success and constraints or limits of possible solutions. Students research and consider multiple possible solutions to a given problem. Generating and testing solutions also becomes more rigorous as the students learn to optimize solutions by revising several times to obtain the best possible design.

Grades 3-5 Engineering Design Standards	
3-5-ETS1-1.	Define a simple design problem reflecting a need or a want that includes specified criteria for success and constraints on materials, time, or cost. *Alignment may include 3-PS2-4, 3-LS4-4, 3-ESS3-1, 4-PS3-4*
3-5-ETS1-2.	Generate and compare multiple possible solutions to a problem based on how well each is likely to meet the criteria and constraints of the problem. *Alignment may include 4-ESS3-2*
3-5-ETS1-3.	Plan and carry out fair tests in which variables are controlled and failure points are considered to identify aspects of a model or prototype that can be improved. *Alignment may include 4-PS4-3*

MS Engineering Design Standards

At the middle school level, students learn to sharpen the focus of problems by precisely specifying criteria and constraints of successful solutions, taking into account not only what needs the problem is intended to meet, but also the larger context within which the problem is defined, including limits to possible solutions. Students can identify elements of different solutions and combine them to create new solutions. Students at this level are expected to use systematic methods to compare different solutions to see which best meet criteria and constraints, and to test and revise solutions a number of times in order to arrive at an optimal design.

Grades 6-8 Engineering Design Standards	
MS-ETS1-1.	Define the criteria and constraints of a design problem with sufficient precision to ensure a successful solution, taking into account relevant scientific principles and potential impacts on people and the natural environment that may limit possible solutions. *Alignment may include MS-PS2-1, MS-PS3-3, MS-LS4-5*
MS-ETS1-2.	Evaluate competing design solutions using a systematic process to determine how well they meet the criteria and constraints of the problem. *Alignment may include MS-PS3-3, MS-LS2-5, MS-ESS3-3*
MS-ETS1-3.	Analyze data from tests to determine similarities and differences among several design solutions to identify the best characteristics of each that can be combined into a new solution to better meet the criteria for success. *Alignment may include MS-PS1-6*

MS-ETS1-4.	Develop a model to generate data for iterative testing and modification of a proposed object, tool, or process such that an optimal design can be achieved. *Alignment may include MS-PS1-6, MS-LS4-5*

HS Engineering Design Standards

Engineering design at the high school level engages students in complex problems that include issues of social and global significance. Such problems need to be broken down into simpler problems to be tackled one at a time. Students are also expected to quantify criteria and constraints so that it will be possible to use quantitative methods to compare the potential of different solutions. While creativity in solving problems is valued, emphasis is on identifying the best solution to a problem, which often involves researching how others have solved it before. Students are expected to use mathematics and/or computer simulations to test solutions under different conditions, prioritize criteria, consider trade-offs, and assess social and environmental impacts.

Grades 9-12 Engineering Design Standards	
HS-ETS1-1.	Analyze a major global challenge to specify qualitative and quantitative criteria and constraints for solutions that account for societal needs and wants. *Alignment may include HS-PS2-3, HS-PS2-6, HS-PS4-2, HS-PS4-5*
HS-ETS1-2.	Design a solution to a complex real-world problem by breaking it down into smaller, more manageable problems that can be solved through engineering. *Alignment may include HS-LS2-7, HS-PS3-3*
HS-ETS1-3.	Evaluate a solution to a complex real-world problem based on prioritized criteria and trade-offs that account for a range of constraints, including cost, safety, reliability, and aesthetics as well as possible social, cultural, and environmental impacts. *Alignment may include HS-PS1-6, HS-PS2-3, HS-LS2-7, HS-LS4-6, HS-ESS3-2, HS-ESS3-4*
HS-ETS1-4.	Use a computer simulation to model the impact of proposed solutions to a complex real-world problem with numerous criteria and constraints on interactions within and between systems relevant to the problem. *Alignment may include HS-LS4-6*